Seloni Sinha

Adm. No. 21JE0855 912-579-3036
selonisinha2003@gmail.com linkedin.com/in/selonisinha selonisinha03

Education

Indian Institute of Technology (IIT) Dhanbad Dec 2021 - May 2025
Bachelor of Technology in Mechanical Engineering (GPA: 8.26) *Dhanbad, Jharkhand*

- **Relevant Coursework:** Object-oriented programming, Data Structures and Algorithms, Building Software systems

Stella Maris Inter College April 2010 - April 2021
*10th Percentage:-99% (**ICSE ALL INDIA RANK 4**), 12th Percentage:-96.25%* *Lucknow, Uttar Pradesh*

Experience

Jaguar Land Rover July 2025 - present
Graduate Software Engineer (Full Time) *Bangalore, Karnataka*

Ontology for Diagnostic Data

- Developing a comprehensive **ontology design** that models key data of vehicle diagnostics.
- Working on facilitating the integration of a **Virtual Knowledge Graph** with a JLR-specific **LLM-powered assistant**.
- Goal is to create a unified ontology design that standardizes and structures vehicle diagnostic data and concepts enabling better **data analysis** across various platforms.

Runbook Automation System

- Working on AI-powered Runbook Automation to automate incident resolution by executing predefined runbooks for triggered control points and error codes.

Jaguar Land Rover May 2024 - July 2024
Data Science Internship (PPO received) *Bangalore, Karnataka*

- Preprocessed a large-scale, multi-source image dataset captured from vehicle-mounted cameras, ensuring consistency in resolution and labeling for the assigned **semantic segmentation** project
- Implemented multiple **deep learning models** aimed at enhancing driver-assist systems and overall user experience, focusing on real-time perception and scene understanding
- Contributed to a research and development project focused on **optimizing fuel efficiency** in vehicles during traffic conditions

Projects

SummarEase | *Python, AWS, GitHub Actions, Docker, FastAPI* *GitHub*

- Developed an end-to-end NLP pipeline for text summarization, leveraging transformers and deep learning models
- Implemented MLOps best practices with GitHub Actions for CI/CD, ensuring automated testing, model retraining, and deployment.
- Deployed the model using Docker, FastAPI and AWS, enabling scalable and efficient API integration
- Monitored model performance with MLflow and logging mechanisms, incorporating feedback loops for continuous improvement

MedIntel | *Python, TensorFlow, MLflow, Docker, DVC* *GitHub*

- Developed a deep learning-based kidney disease classification model using CNNs and traditional ML algorithms for accurate diagnosis
- Implemented MLOps best practices with MLflow for experiment tracking and DVC for dataset versioning, ensuring reproducibility
- Deployed the model using FastAPI, Docker, and AWS to enable real-time inference
- Automated CI/CD workflows with GitHub Actions, optimizing model retraining and deployment for continuous improvement

RAGenius | *Python, Llama2, FAISS, AWS Bedrock, LangChain, Boto3* *GitHub*

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system using Llama 2, AWS Bedrock, and LangChain for intelligent and context-aware responses
- Integrated LLM with FAISS to enable efficient document retrieval and intelligent response generation
- Built a serverless and scalable architecture using AWS Lambda, API Gateway, DynamoDB, and S3, leveraging Boto3 for AWS service interactions
- Automated MLOps workflows with GitHub Actions and AWS SageMaker, enabling continuous model improvement and seamless deployment

Technical Skills

Languages: Python, C, C++, Javascript
Web Development: HTML, CSS, Bootstrap, NodeJS, ExpressJS, ReactJS
ML Frameworks/Libraries: PyTorch, TensorFlow, Owlready, RDFLib, Numpy, Pandas, Matplotlib, Scikit-learn, FastAPI, Hugging Face Transformers, Amazon SageMaker, RMLMapper
Database: MongoDB, SQL, Redis
Tools: GitHub, GitLab, WebVOWL, Ontop, Postman, VSCode, Jupyter notebook, Kiro
Concepts: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Agentic AI, Model Context Protocol(MCP), LangChain, LangGraph, Ontologies and Knowledge Graphs, Web Ontology Language (OWL), Domain Specific Language (DSL), Natural Language Processing, Object Oriented Programming, Data Structures and Algorithms, Competitive Programming

Achievements

- **All India Rank 4** in ICSE Class 10 Board Examination among 2 million candidates
- **Top 1%** All India in JEE national exam
- **1st Rank** in National EV Thon competition in east India's largest tech fest
- Solved 400+ problems across all coding platforms
- Secured Global Rank 593 in Codechef Starters
- Codechef rating - 1455, Codeforces rating- 1095
- Hackerrank - 5 star

Social Engagements

Organizer at HackFest'23: 36-hour long official hackathon organised by IIT (ISM) Dhanbad
Technical Team Associate : Steering and Suspension division at Electric Mobility Club, the official Formula Racing club of IIT (ISM) Dhanbad
Operations and Sponsorship team member : At Electric Mobility Club
Active member at CODEISM: The coding club of IIT Dhanbad
School Representative : Inter School Literary Competition